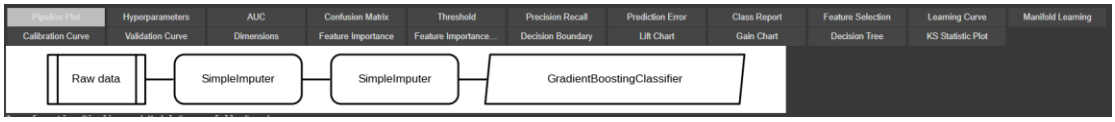
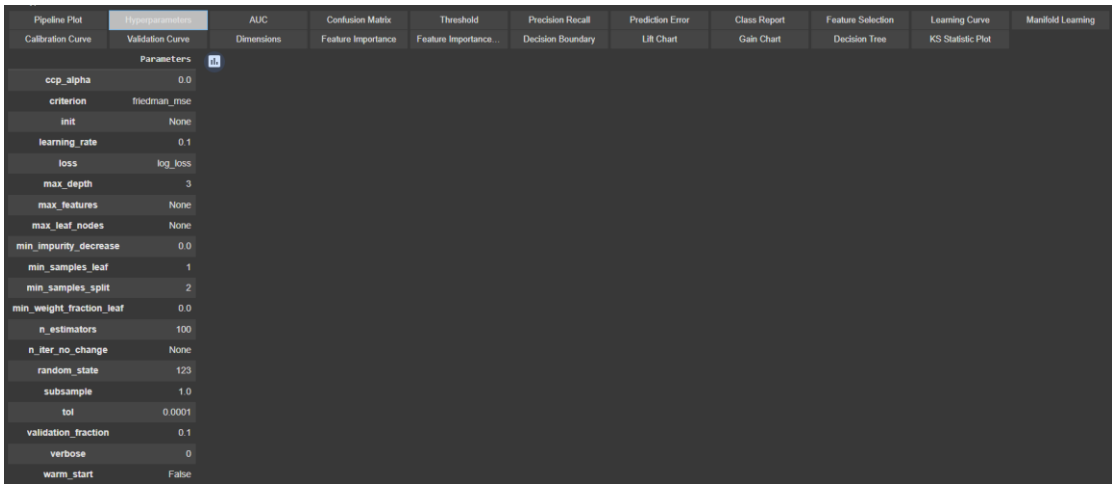
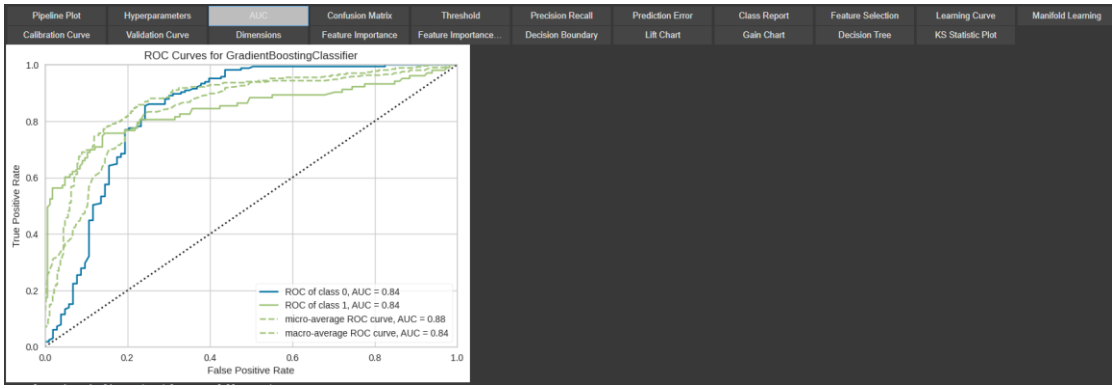
HW4-1

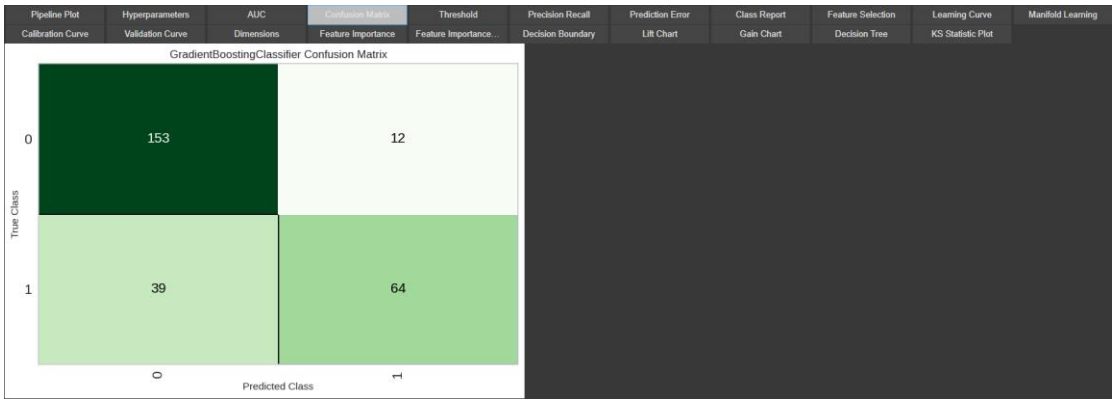
	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
gbc	Gradient Boosting Classifier	0.8265	0.8438	0.7111	0.8141	0.7571	0.6235	0.6286	0.1880
rf	Random Forest Classifier	0.8041	0.8446	0.7109	0.7681	0.7336	0.5796	0.5848	0.1900
lightgbm	Light Gradient Boosting Machine	0.8009	0.8366	0.7067	0.7594	0.7298	0.5728	0.5758	0.2710
lr	Logistic Regression	0.7976	0.8507	0.7197	0.7470	0.7300	0.5688	0.5719	0.4700
ada	Ada Boost Classifier	0.7959	0.8424	0.7241	0.7433	0.7303	0.5668	0.5699	0.1080
xgboost	Extreme Gradient Boosting	0.7912	0.8294	0.6942	0.7447	0.7153	0.5514	0.5550	0.0680
ridge	Ridge Classifier	0.7880	0.8505	0.6987	0.7365	0.7146	0.5466	0.5492	0.0320
lda	Linear Discriminant Analysis	0.7880	0.8504	0.6987	0.7365	0.7146	0.5466	0.5492	0.0390
et	Extra Trees Classifier	0.7848	0.8216	0.6980	0.7343	0.7120	0.5407	0.5443	0.2530
qda	Quadratic Discriminant Analysis	0.7833	0.8454	0.6402	0.7658	0.6898	0.5267	0.5376	0.0280
nb	Naive Bayes	0.7831	0.8305	0.7324	0.7153	0.7197	0.5435	0.5480	0.0270
dt	Decision Tree Classifier	0.7496	0.7243	0.6569	0.6813	0.6667	0.4666	0.4687	0.0310
knn	K Neighbors Classifier	0.6773	0.7294	0.5319	0.5899	0.5532	0.3041	0.3084	0.0430
svm	SVM - Linear Kernel	0.6740	0.7257	0.6786	0.5391	0.5823	0.3362	0.3595	0.0270
dummy	Dummy Classifier	0.6164	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0250

```
Best Model: GradientBoostingClassifier(ccp_alpha=0.0, criterion='friedman_mse', init=None,
                           learning_rate=0.1, loss='log_loss', max_depth=3,
                           max_features=None, max_leaf_nodes=None,
                           min_impurity_decrease=0.0, min_samples_leaf=1,
                           min_samples_split=2, min_weight_fraction_leaf=0.0,
                           n_estimators=100, n_iter_no_change=None,
                           random_state=123, subsample=1.0, tol=0.0001,
                           validation_fraction=0.1, verbose=0,
                           warm_start=False)
```






Pipeline Plot
Hyperparameters
AUC
Confusion Matrix
Threshold
Precision Recall
Prediction Error
Class Report
Feature Selection
Learning Curve
Manifold Learning
Calibration Curve
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
ROC Curves for GradientBoostingClassifier
True Positive Rate
False Positive Rate
ROC of class 0, AUC = 0.84
ROC of class 1, AUC = 0.84
micro-average ROC curve, AUC = 0.88
macro-average ROC curve, AUC = 0.84


Pipeline Plot
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Feature Importance...
Decision Boundary
Lift Chart
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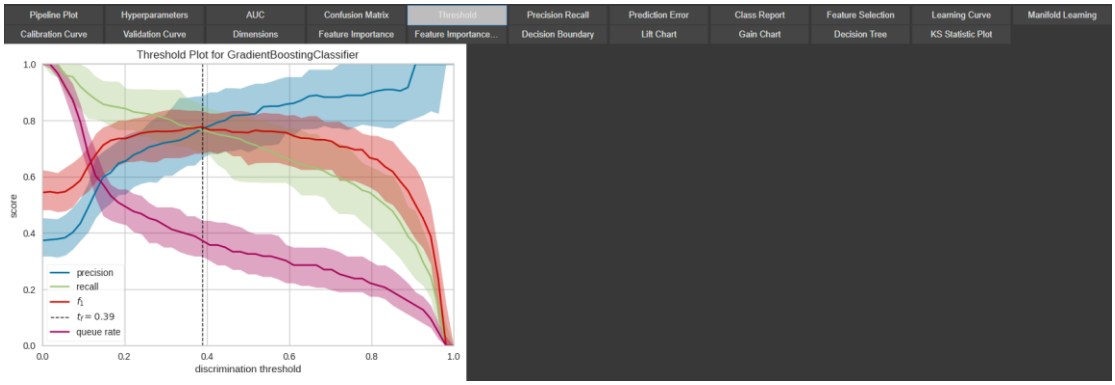
Decision Tree
KS Statistic Plot
GradientBoostingClassifier Confusion Matrix
True Class
Predicted Class
153
12
39
64


Pipeline Plot
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Feature Importance...
Decision Boundary
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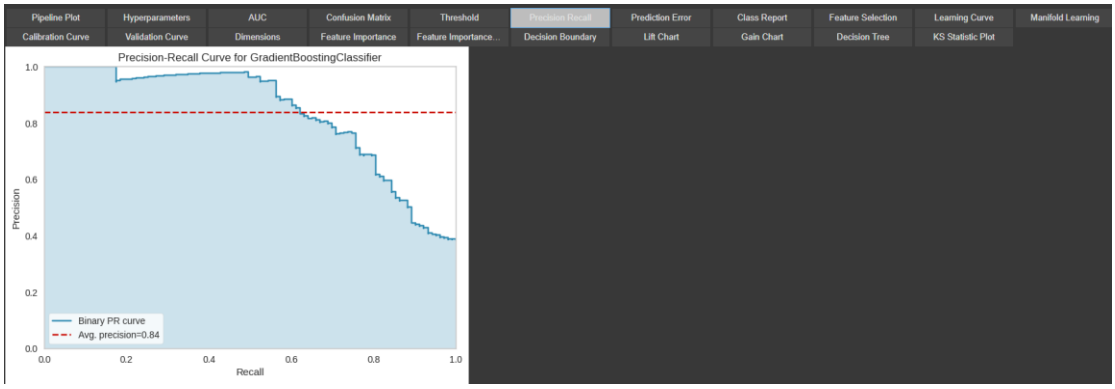
Gain Chart
Decision Tree
KS Statistic Plot
Threshold Plot for GradientBoostingClassifier
score
discrimination threshold
precision
recall
f_1
$t_f = 0.39$
queue rate


Pipeline Plot
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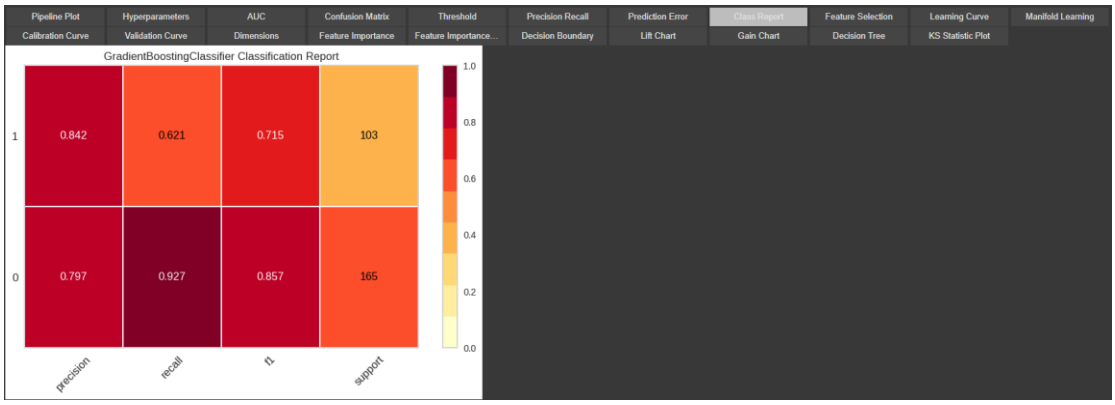
Gain Chart
Decision Tree
KS Statistic Plot
Precision-Recall Curve for GradientBoostingClassifier
Precision
Recall
Binary PR curve
Avg. precision=0.84


Pipeline Plot
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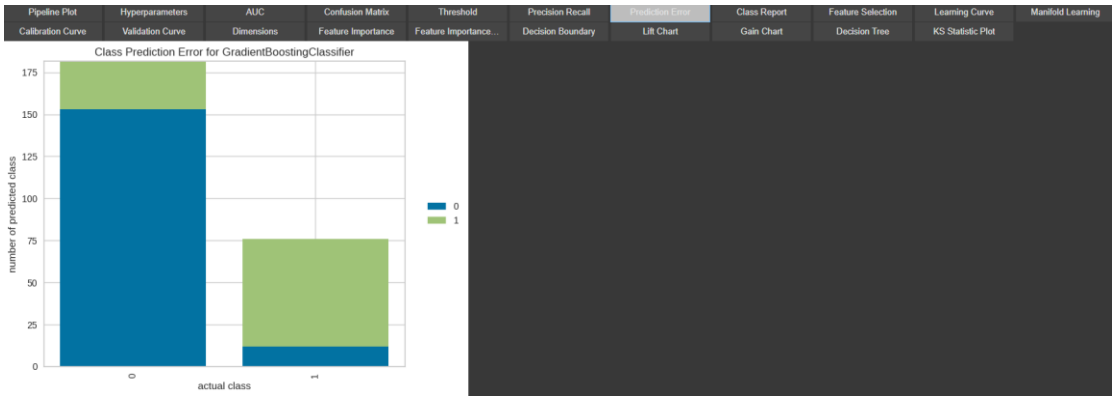
KS Statistic Plot
GradientBoostingClassifier Classification Report
1
0
0.842
0.621
0.715
103
0.797
0.927
0.857
165
precision
recall
f1
support
1.0
0.8
0.6
0.4
0.2
0.0


Pipeline Plot
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Dimensions
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Feature Importance...
Decision Boundary
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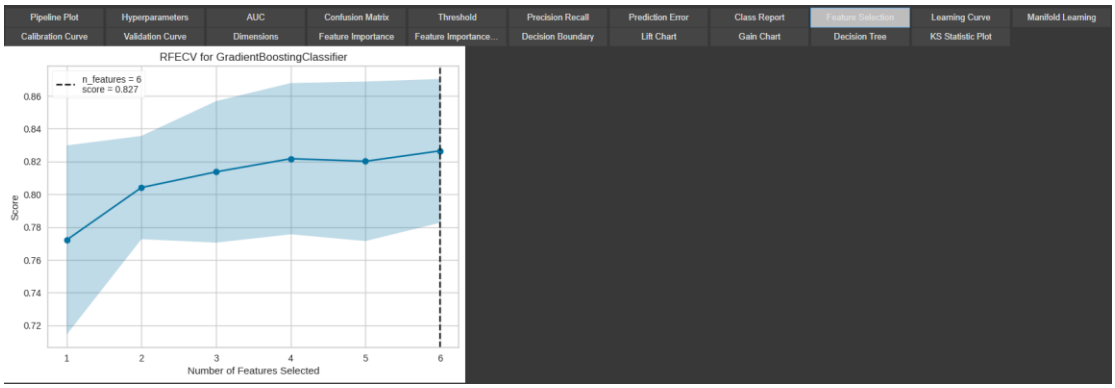
Decision Tree
KS Statistic Plot
Class Prediction Error for GradientBoostingClassifier
number of predicted class
actual class
0
1
175
150
125
100
75
50
25
0


Pipeline Plot
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Confusion Matrix
Threshold
Precision Recall
Prediction Error
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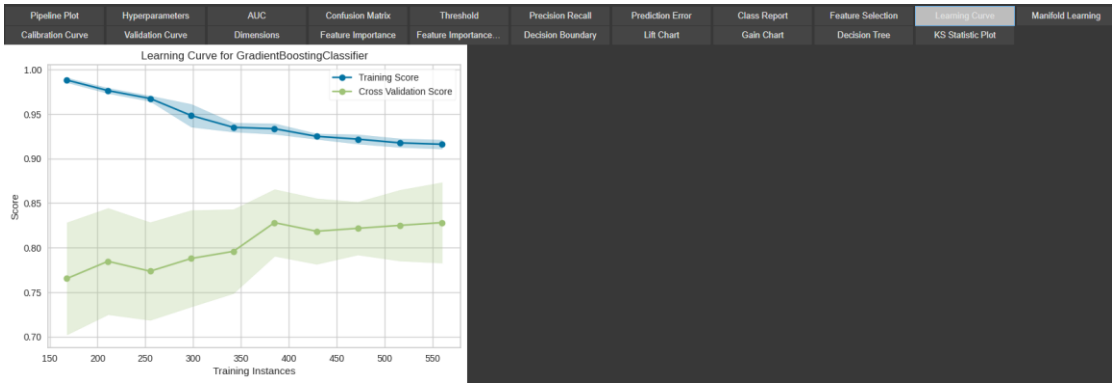
Manifold Learning
Calibration Curve
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
RFECV for GradientBoostingClassifier
n_features = 6
score = 0.827
Score
Number of Features Selected
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.72
1
2
3
4
5
6


Pipeline Plot
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Manifold Learning
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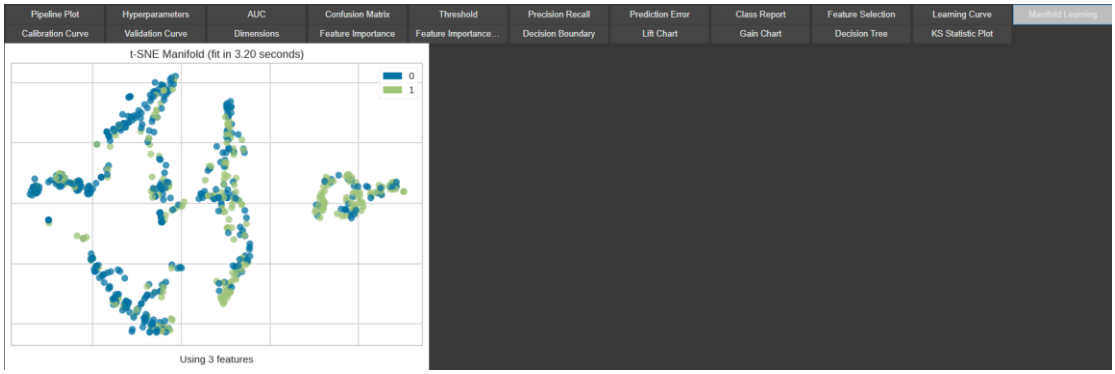
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
Learning Curve for GradientBoostingClassifier
Training Score
Cross Validation Score
Score
Training Instances
1.00
0.95
0.90
0.85
0.80
0.75
0.70
150
200
250
300
350
400
450
500
550


Pipeline Plot
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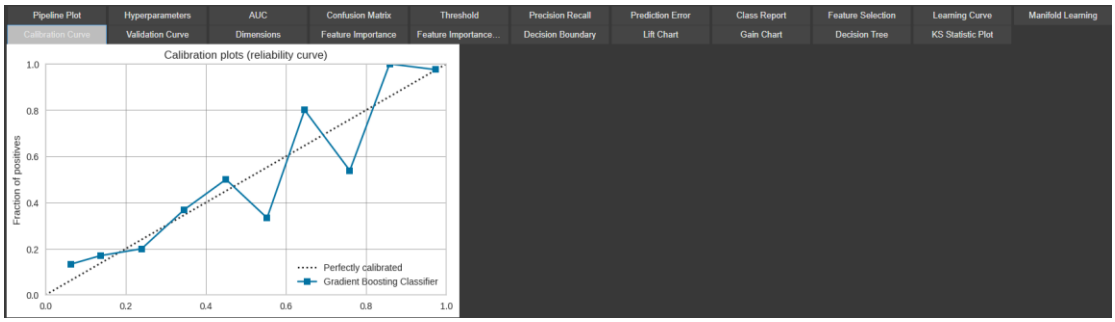
Learning Curve
Manifold Learning
Calibration Curve
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
t-SNE Manifold (fit in 3.20 seconds)
0
1
Using 3 features


Pipeline Plot
Hyperparameters
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Confusion Matrix
Threshold
Precision Recall
Prediction Error
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Learning Curve
Manifold Learning
Calibration Curve
Validation Curve
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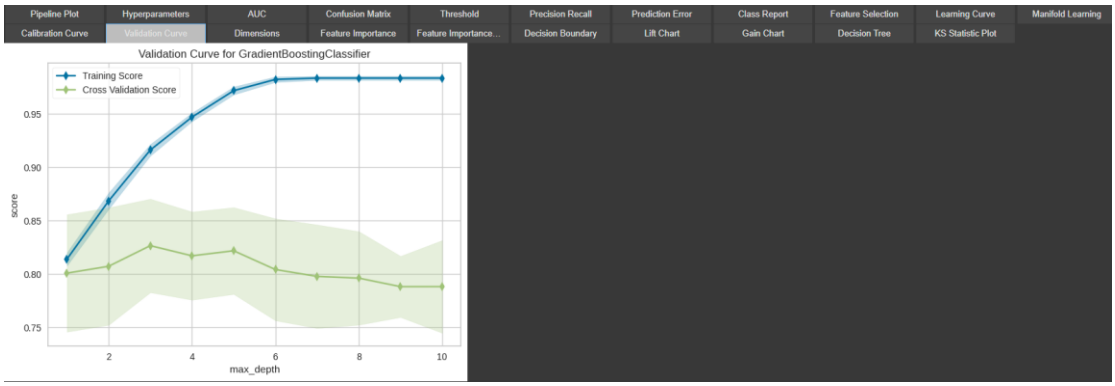
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
Calibration plots (reliability curve)
Fraction of positives
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Perfectly calibrated
Gradient Boosting Classifier


Pipeline Plot
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Confusion Matrix
Threshold
Precision Recall
Prediction Error
Class Report
Feature Selection
Learning Curve
Manifold Learning
Calibration Curve
Validation Curve
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Feature Importance...
Decision Boundary
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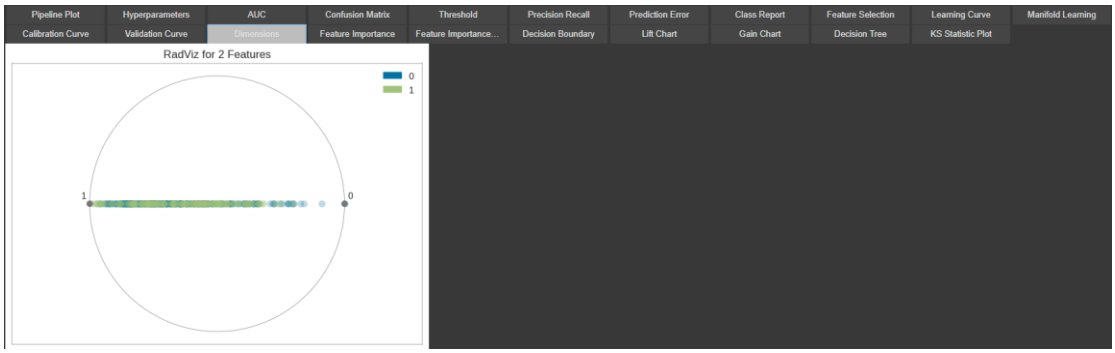
Validation Curve for GradientBoostingClassifier
Training Score
Cross Validation Score
score
0.95
0.90
0.85
0.80
0.75
2
4
6
8
10
max_depth


Pipeline Plot
Hyperparameters
AUC
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Prediction Error
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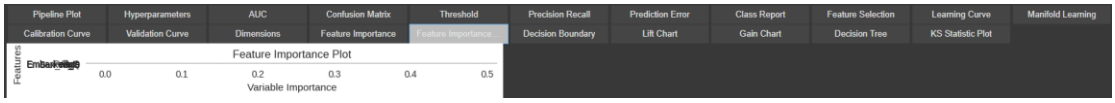
Calibration Curve
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
RadViz for 2 Features
0
1
1
0


Pipeline Plot
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Learning Curve
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Calibration Curve
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
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KS Statistic Plot
Feature Importance Plot
Features
0.0
0.1
0.2
0.3
0.4
0.5
Variable Importance

Pipeline Plot
Hyperparameters
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Confusion Matrix
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Precision Recall
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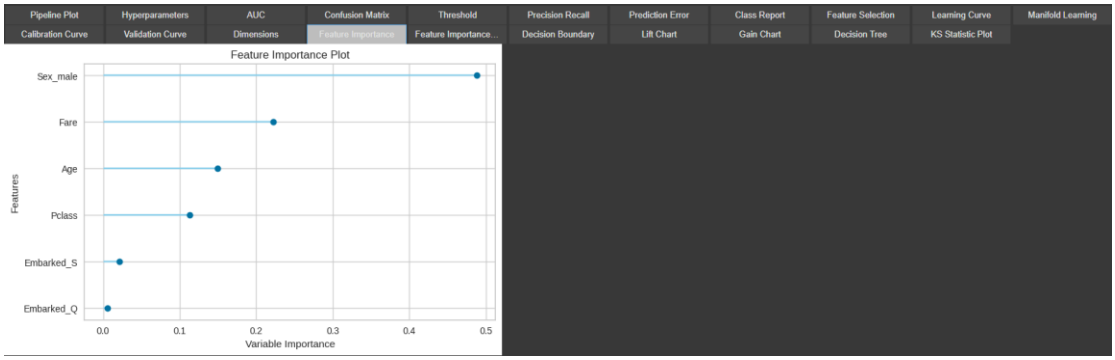
Validation Curve
Dimensions
Feature Importance
Feature Importance...
Decision Boundary
Lift Chart
Gain Chart
Decision Tree
KS Statistic Plot
0
1
Feature Two
Feature One


Pipeline Plot
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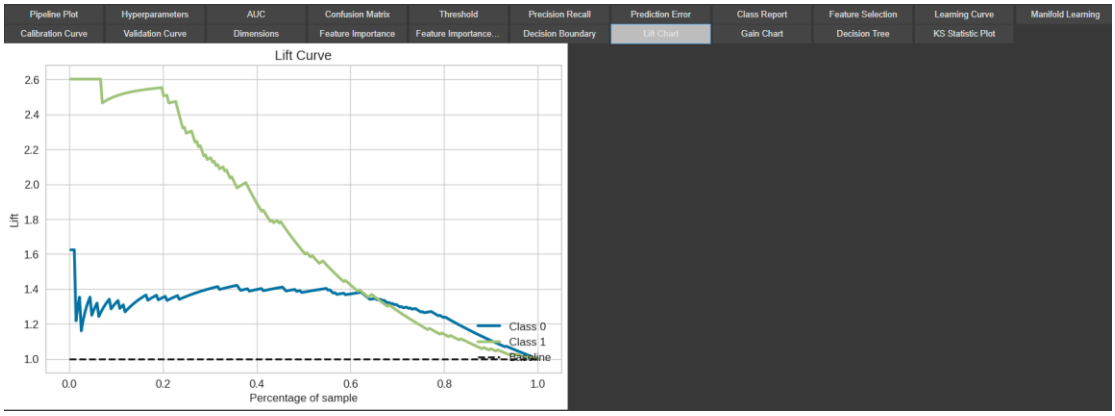
Decision Tree
KS Statistic Plot
Feature Importance Plot
Sex_male
Fare
Age
Pclass
Embarked_S
Embarked_Q
Features
Variable Importance


Pipeline Plot
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Feature Importance...
Decision Boundary
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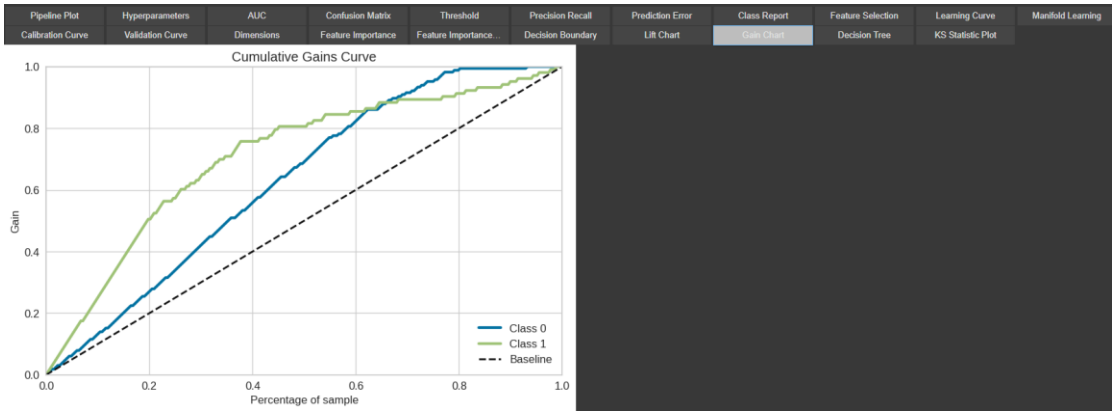
Gain Chart
Decision Tree
KS Statistic Plot
Lift Curve
Lift
Class 0
Class 1
Baseline
Percentage of sample


Pipeline Plot
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AUC
Confusion Matrix
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Precision Recall
Prediction Error
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Learning Curve
Manifold Learning
Calibration Curve
Validation Curve
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Feature Importance...
Decision Boundary
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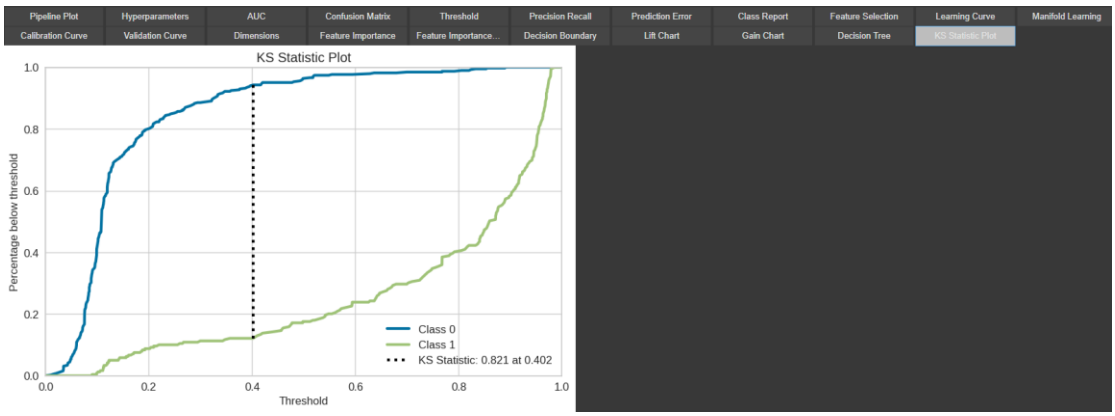
Decision Tree
KS Statistic Plot
Cumulative Gains Curve
Gain
Class 0
Class 1
Baseline
Percentage of sample


Pipeline Plot
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KS Statistic Plot
Percentage below threshold
Threshold
0.0
0.2
0.4
0.6
0.8
1.0
Class 0
Class 1
KS Statistic: 0.821 at 0.402